

CH AAKSHAY

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PROFESSIONAL SUMMARY

Adaptable **Electronics & Communication Engineering** student (B.Tech, PES University, Class of 2027) with a **client-focused approach** to translating ambiguous operational and business requirements into reliable technical solutions. Experienced in building automated data processing pipelines with **Python, Pandas, and Microsoft Excel**, and developing full-stack web applications with **Next.js and TypeScript**. Proven track record of collaborating across engineering teams, internal client groups, and operations stakeholders, backed by structured testing, technical documentation, and measurable delivery outcomes.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C/C++, SQL, Bash, Git, RESTful APIs

Data Analytics & Tools: Pandas, NumPy, Microsoft Excel, Data Cleaning & ETL Pipelines, Outlier Detection (IQR), Time-Series Ingestion, Statistical Modeling, Plotly

Application Development & Web: Next.js 14, React, Node.js, JSON, HTML5/CSS3, Tailwind CSS, Responsive Web Design

Security & Risk Controls: Cyber Fraud Incident Workflows, Data Privacy Controls, Access Validation, OCR Integration, Automated Document Generation

System Integration & Infrastructure: OPC-UA, Modbus TCP, Industrial SCADA Telemetry, Microservices Architecture, Linux Environment

Testing & Collaboration: Automated Unit Testing, Test Data Preparation, Accessibility Testing (axe-core), Version Control (Git/GitHub), Agile/Scrum

PROFESSIONAL EXPERIENCE

Industrial Technology & Data Intern — NoPo Nanotechnologies India Pvt. Ltd.

Jun 2026 – Aug 2026 | Bengaluru, India

- **Client-Focused Delivery & Stakeholder Collaboration:** Partnered directly with plant operations and engineering teams as an internal client group to gather functional requirements, resolving data discrepancies and ensuring automated reports met downstream client quality specifications.
- **Large-Scale Data Analytics & Pipeline Engineering:** Engineered an automated Python-based data ingestion and transformation pipeline (`reactor_to_plc_generic.py`) processing time-series telemetry across 6 industrial reactors (66 sensor tags including thermocouples, mass flow, and pressure).
- **Data Quality & Risk Controls:** Implemented automated validation rules using Interquartile Range (IQR) outlier detection and sensor-fault monitoring, eliminating manual data cleansing and ensuring 99%+ operational logging integrity.
- **Process Optimization & System Integration:** Automated the conversion of high-frequency OPC historian logs into standardized, multi-sheet Excel reports (*PLC Data, Flags, Summary*), reducing daily reporting turnaround time from 2 hours to under 30 seconds.
- **Technical Documentation & Reliability:** Authored operational runbooks and data schemas; conducted code unit testing and edge-case validation against multi-megabyte industrial logs to ensure zero-loss data persistence.

KEY PROJECTS

Cyber Incident Response & Risk Mitigation Platform (CyberRakshak)

Next.js 14, TypeScript, OCR (Tesseract.js), jsPDF, axe-core

- **Solution Architecture:** Architected a full-stack emergency response web application to guide victims of financial cyber fraud through statutory restitution workflows.
- **Automated Document Generation:** Integrated OCR (Tesseract.js) to parse transaction screenshots, automatically drafting standardized police FIR complaint dossiers, Section 503 restitution petitions, and bank freeze notices in PDF.
- **Testing & Reliability:** Implemented automated unit test suites (`test_runner.mjs`) and accessibility compliance tests (axe-core), ensuring high reliability, zero runtime crashes, and seamless usability under stress.

Real-Time Geospatial Incident Platform & Dispatch Hub (SafeMate)

Next.js 14, TypeScript, REST APIs, Leaflet, Web Audio API

- **Multi-Source Data Fusion:** Developed an event-driven crisis monitoring platform integrating 4 public data streams (UN GDACS disaster telemetry, weather, news, and OpenStreetMap).
- **Proximity Detection & Alerting:** Built a real-time proximity-detection engine delivering danger-zone breach alerts in under 100ms, paired with automated emergency dispatch signaling.
- **Testing & Deployment:** Executed end-to-end integration and user acceptance testing across simulated network dropouts and high-frequency GPS telemetry streams.

Financial Modeling & Risk Analytics Platform (FinPal / FinCal)

Python, Streamlit, Pandas, NumPy, Plotly

- **Financial Decision Modeling:** Developed an interactive quantitative decision-support tool modeling asset compounding, cash flow projections, and regulatory tax calculations (LTCG).
- **Scenario Analysis:** Programmed a Monte Carlo simulation engine evaluating portfolio return variance and inflation impacts across multi-decade tenures, presenting clear visual insights via Plotly.

EDUCATION

PES University, Bengaluru, Karnataka

2023 – Expected 2027

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering — **CGPA: 6.61 / 10.0**

Relevant Coursework: Embedded Microcontroller Systems, Computer Architecture & Organization, Digital Signal Processing (DSP), Signals & Systems, Probability & Statistics, Linear Algebra.

BASE PU College, Bengaluru — Pre-University Certificate (PUC II): **86.2%**

2023

Innisfree House School, Bengaluru — ICSE Class X: **89.8%**

2021

PROFESSIONAL ATTRIBUTES & AVAILABILITY

Core Competencies: Client-Focused Communication, Business Requirements Translation, Cross-Functional Teamwork, Technical & Process Documentation, Analytical Reasoning.

Workplace Adaptability: Ready and willing to work in flexible shift timings, support client deliverables, and collaborate from Offshore Delivery Centers (ODC) or client locations as required.

Languages: English (Professional Proficiency), Hindi (Professional Proficiency), Kannada (Native).